VOLUME II

ANSWERS

UNIT 8

MCQ

1. $Ag_2C_2O_4 \rightleftharpoons 2Ag^+ + C_2O_4^{2-}$

$[Ag^+] = 2.24\times10^{-4} \text{ mol } L^{-1}$

$[C_2O_4^{2-}] = \frac{2.24\times10^{-4}}{2} \text{ mol } L^{-1}$

$= 1.12\times10^{-4} \text{ mol } L^{-1}$

$K_{sp} = [Ag^+]^2 [C_2O_4^{2-}]$

$= (2.24\times10^{-4} \text{ mol } L^{-1})^2 (1.12\times10^{-4} \text{ mol } L^{-1})$

$= 5.619\times10^{-12} \text{ mol}^3 L^{-3}$

[Option (d)]

2. iii) 75 ml $\frac{M}{5}$ HCl + 25ml $\frac{M}{5}$ NaOH

No of moles of HCl = $0.2\times75\times10^{-3} = 15 \times 10^{-3}$

No of moles of NaOH = $0.2 \times 25 \times 10^{-3} = 5 \times 10^{-3}$

No of moles of HCl after mixing = $15 \times 10^{-3} - 5 \times 10^{-3}$

$= 10 \times 10^{-3}$

$\therefore$ concentration of HCl = $\frac{\text{No of moles of HCl}}{\text{Vol in litre}}$

$= \frac{10\times10^{-3}}{100\times10^{-3}} = 0.1M$

for (iii) solution, pH of 0.1M HCl = $-\log_{10}(0.1)$

$= 1.$

[Option (d)].

3. $BaSO_4 \rightleftharpoons Ba^{2+} + SO_4^{2-}$

$K_{sp} = (s)\ (s)$

$K_{sp} = (s)^2$

$= (2.42\times10^{-3} \text{ g } L^{-1})^2$

$= \left(\frac{2.42\times10^{-3} \text{ g } L^{-1}}{233 \text{g mol}^{-1}}\right)^2$

$= (0.01038\times10^{-3})^2$

$= (1.038\times10^{-5})^2$

$= 1.077\times10^{-10}$

$= 1.08\times10^{-10} \text{ mol}^2 \text{ } L^{-2}$

[Option (c)]

4. $Ca(OH)_2 \rightleftharpoons Ca^{2+} + 2OH^-$

Given that pH=9

pOH=14-9=5

$[pOH = -\log_{10}[OH^-]]$

$\therefore [OH^-] = 10^{-pOH}$

$[OH^-] = 10^{-5} \text{ M}$

$K_{sp} = [Ca^{2+}][OH^-]^2$

$= \frac{10^{-5}}{2} \times (10^{-5})^2$

$= 0.5\times10^{-15}$

[Option (a)]

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